

Chapter 7

Martingales

Martingales play a central role in the modern theory of stochastic processes and stochastic calculus. Martingales constructed from a Brownian motion were considered in Section 3.3 and martingales arising in diffusions in Section 6.1. Martingales have a constant expectation, which remains the same under random stopping. Martingales converge almost surely. Stochastic integrals are martingales. These are the most important properties of martingales, which hold under some conditions.

7.1 Definitions

The main ingredient in the definition of a martingale is the concept of conditional expectation, consult Chapter 2 for its definition and properties.

Definition 7.1 *A stochastic process $M(t)$, where time t is continuous $0 \leq t \leq T$, or discrete $t = 0, 1, \dots, T$, adapted to a filtration $\mathbb{F} = (\mathcal{F}_t)$ is a martingale if for any t , $M(t)$ is integrable, that is, $E|M(t)| < \infty$ and for any t and s with $0 \leq s < t \leq T$,*

$$E(M(t)|\mathcal{F}_s) = M(s) \quad a.s. \quad (7.1)$$

$M(t)$ is a martingale on $[0, \infty)$ if it is integrable and the martingale property (7.1) holds for any $0 \leq s < t < \infty$.

Definition 7.2 *A stochastic process $X(t)$, $t \geq 0$ adapted to a filtration $\mathbb{F}$ is a supermartingale (submartingale) if it is integrable, and for any t and s , $0 \leq s < t \leq T$*

$$E(X(t)|\mathcal{F}_s) \leq X(s), \quad (E(X(t)|\mathcal{F}_s) \geq X(s)) \quad a.s.$$

If $X(t)$ is a supermartingale, then $-X(t)$ is a submartingale. The mean of a supermartingale is non-increasing with t , the mean of a submartingale is non-decreasing in t , and the mean of a martingale is constant in t . This property is used in a test for a super(sub)martingale to be a true martingale.

Theorem 7.3 *A supermartingale $M(t)$, $0 \leq t \leq T$, is a martingale if and only if $EM(T) = EM(0)$.*

PROOF: If M is a martingale, then $EM(T) = EM(0)$ follows by the martingale property with $s = 0$ and $t = T$. Conversely, suppose $M(t)$ is a supermartingale and $EM(T) = EM(0)$. If for some t and s we have a strict inequality, $E(M(t)|\mathcal{F}_s) < M(s)$ on a set of positive probability, then by taking expectations, we obtain $EM(t) < EM(s)$. Since the expectation of a supermartingale is non-increasing, $EM(T) \leq EM(t) < EM(s) \leq EM(0)$. But this contradicts the condition of the theorem $EM(T) = EM(0)$. Thus for all t and s the inequality $E(M(t)|\mathcal{F}_s) \leq M(s)$ must be an equality almost surely. $\square$

We refer to Theorem 2.32 on the existence of the regular right-continuous version for supermartingales. Regular right-continuous versions of processes will be taken.

Square Integrable Martingales

A special role in the theory of integration is played by square integrable martingales.

Definition 7.4 *A random variable X is square integrable if $E(X^2) < \infty$. A process $X(t)$ on the time interval $[0, T]$, where T can be infinite, is square integrable if $\sup_{t \in [0, T]} EX^2(t) < \infty$ ($\sup_{t \geq 0} EX^2(t) < \infty$), i.e. second moments are bounded.*

Example 7.1:

1. Brownian motion $B(t)$ on a finite time interval $0 \leq t \leq T$ is a square integrable martingale, since $EB^2(t) = t < T < \infty$. Similarly, $B^2(t) - t$ is a square integrable martingale. They are not square integrable when $T = \infty$.
2. If $f(x)$ is bounded and continuous function on $\mathbb{R}$, then Itô integrals $\int_0^t f(B(s))dB(s)$ and $\int_0^t f(s)dB(s)$ are square integrable martingales on any finite time interval $0 \leq t \leq T$. Indeed, by (4.7), an Itô integral is a martingale, and since $|f(x)| \leq K$,

$$E \left(\int_0^t f(B(s))dB(s) \right)^2 = E \left(\int_0^t f^2(B(s))ds \right) \leq K^2 t \leq K^2 T < \infty.$$

If moreover, $\int_0^\infty f^2(s)ds < \infty$ then $\int_0^t f(s)dB(s)$ is a square integrable martingale on $[0, \infty)$.

7.2 Uniform Integrability

To appreciate the definition of uniform integrability of a process, recall what is meant by integrability of a random variable X . It is called integrable if $E|X| < \infty$. It is easy to see that this holds if and only if

$$\lim_{n \rightarrow \infty} E(|X|I(|X| > n)) = 0. \quad (7.2)$$

Indeed, if X is integrable then (7.2) holds by the dominated convergence, since $\lim_{n \rightarrow \infty} |X|I(|X| > n) = 0$ and $|X|I(|X| > n) \leq |X|$. Conversely, let n be large enough for the rhs in (7.2) to be finite. Then

$$E|X| = E(|X|I(|X| > n)) + E(|X|I(|X| \leq n)) < \infty,$$

since the first term is finite by (7.2) and the second is bounded by n .

Definition 7.5 A process $X(t)$, $0 \leq t \leq T$ is called uniformly integrable if $E(|X(t)|I(|X(t)| > n))$ converges to zero as $n \rightarrow \infty$ uniformly in t , that is,

$$\lim_{n \rightarrow \infty} \sup_t E(|X(t)|I(|X(t)| > n)) = 0, \quad (7.3)$$

where the supremum is over $[0, T]$ in the case of a finite time interval and $[0, \infty)$ if the process is considered on $0 \leq t < \infty$.

Example 7.2: We show that if $X(t)$, $0 \leq t \leq T$ is uniformly integrable, then it is integrable, that is, $\sup_t E|X(t)| < \infty$. Indeed,

$$\sup_t E|X(t)| < \sup_t E(|X(t)|I(|X(t)| > n)) + n.$$

Since $X(t)$ is uniformly integrable, the first term converges to zero as $n \rightarrow \infty$, in particular it is bounded, and the result follows.

Sufficient conditions for uniform integrability are given next.

Theorem 7.6 If the process X is dominated by an integrable random variable, $|X(t)| \leq Y$ and $E(Y) < \infty$, then it is uniformly integrable. In particular, if $E(\sup_t |X(t)|) < \infty$, then it is uniformly integrable.

PROOF: $E(|X(t)|I(|X(t)| > n)) < E(|Y|I(|Y| > n)) \rightarrow 0$, as $n \rightarrow \infty$. □

Note that there are uniformly integrable processes (martingales) which are not dominated by an integrable random variable, so that the sufficient condition for uniform integrability $E(\sup_t |X(t)|) < \infty$ is not necessary for uniform integrability. Another sufficient condition for uniform integrability is given by the following result, see for example Protter (1992), p.9, Liptser and Shiryaev (2001), p. 17.

Theorem 7.7 *If for some positive, increasing, convex function $G(x)$ on $[0, \infty)$ such that $\lim_{x \rightarrow \infty} G(x)/x = \infty$,*

$$\sup_{t \leq T} E(G(|X(t)|)) < \infty, \quad (7.4)$$

then $X(t)$, $t \leq T$ is uniformly integrable.

We omit the proof. In practice the above result is used with $G(x) = x^r$ for $r > 1$, and uniform integrability is checked by using moments. For second moments $r = 2$, we have: square integrability implies uniform integrability.

Corollary 7.8 *If $X(t)$ is square integrable, that is, $\sup_t EX^2(t) < \infty$, then it is uniformly integrable.*

In view of this, examples of uniformly integrable martingales are provided by square integrable martingales given in Example 7.1. The following result provides a construction of uniformly integrable martingales.

Theorem 7.9 (Doob's, Levy's martingale) *Let Y be an integrable random variable, that is, $E|Y| < \infty$ and define*

$$M(t) = E(Y|\mathcal{F}_t). \quad (7.5)$$

Then $M(t)$ is a uniformly integrable martingale.

PROOF: It is easy to see that $M(t)$ is a martingale. Indeed, by the law of double expectation, $E(M(t)|\mathcal{F}_s) = E(E(Y|\mathcal{F}_t)|\mathcal{F}_s) = E(Y|\mathcal{F}_s) = M(s)$. The proof of uniform integrability is more involved. It is enough to establish the result for $Y \geq 0$ as the general case will follow by considering Y^+ and Y^- . If $Y \geq 0$ then $M(t) \geq 0$ for all t . We show next that $M^* = \sup_{t \leq T} M(t) < \infty$. If not, there is a sequence of $t_n \uparrow \infty$ such that $M(t_n) \uparrow \infty$. By monotone convergence, $EM(t_n) \uparrow \infty$, which is a contradiction, as $EM(t_n) = EY < \infty$. Now, by the general definition of conditional expectation, see (2.16), $E(M(t)I(M(t) > n)) = E(YI(M(t) > n))$. Since $\{M(t) > n\} \subseteq \{M^* > n\}$, $E(YI(M(t) > n)) \leq E(YI(M^* > n))$. Thus $E(M(t)I(M(t) > n)) \leq E(YI(M^* > n))$. Since the right-hand side does not depend on t , $\sup_{t \leq T} E(M(t)I(M(t) > n)) \leq E(YI(M^* > n))$. But this converges to zero as $n \rightarrow \infty$, because M^* is finite and Y is integrable. $\square$

The martingale in (7.5) is said to be *closed* by Y . An immediate corollary is

Corollary 7.10 *Any martingale $M(t)$ on a finite time interval $0 \leq t \leq T < \infty$ is uniformly integrable and is closed by $M(T)$.*

It will be seen in the next section that a uniformly integrable martingale on $[0, \infty)$ is also of the form (7.5). That is, there exists a random variable, called $M(\infty)$ such that the martingale property holds for all $0 \leq s < t$, including $t = \infty$.

7.3 Martingale Convergence

In this section martingales on the infinite time interval $[0, \infty)$ are considered.

Theorem 7.11 (Martingale Convergence Theorem) *If $M(t)$, $0 \leq t < \infty$, is an integrable martingale (supermartingale or submartingale), that is, if $\sup_{t \geq 0} E|M(t)| < \infty$, then there exists an almost sure limit $\lim_{t \rightarrow \infty} M(t) = Y$ and Y is an integrable random variable.*

The proof of this result is due to Doob and it is too involved to be given here. If $M(t)$ is a martingale, then the condition $\sup_{t \geq 0} E|M(t)| < \infty$ is equivalent to any of the following conditions:

- $\lim_{t \rightarrow \infty} E|M(t)| < \infty$. This is because $|x|$ is a convex function, implying that $|M(t)|$ is a submartingale, and expectation of a submartingale is an increasing function of t . Hence the supremum is the same as the limit.
- $\lim_{t \rightarrow \infty} EM^+(t) < \infty$. This is because $E|M(t)| = EM^+(t) + EM^-(t)$. If $EM(t) = c$, then $EM(t) = EM^+(t) - EM^-(t) = c$ and $EM^+(t) = EM^-(t) + c$.
- $\lim_{t \rightarrow \infty} EM^-(t) < \infty$.

If $M(t)$ is a submartingale, it is enough to demand $\sup_t EM^+(t) < \infty$, and if it is a supermartingale it is enough to demand $\sup_t EM^-(t) < \infty$, for the existence of a finite limit.

Corollary 7.12

1. *Uniformly integrable martingales converge almost surely.*
2. *Square integrable martingales converge almost surely.*
3. *Positive martingales converge almost surely.*
4. *Submartingales bounded from above (negative) converge almost surely.*
5. *Supermartingales bounded from below (positive) converge almost surely.*

PROOF: Since uniformly integrable martingales are integrable, they converge. Since square integrable martingales are uniformly integrable, they converge. If $M(t)$ is positive then $|M(t)| = M(t)$, and $E|M(t)| = EM(t) = EM(0) < \infty$. □

Note that expectations $EM(t)$ may or may not converge to the expectation of the limit EY (see Example 7.3). The case when $EY = \lim_{t \rightarrow \infty} EM(t)$, is precisely when $M(t)$ is uniformly integrable. The next result, given without proof, shows that uniformly integrable martingales have form (7.5).

Theorem 7.13 *If $M(t)$ is a uniformly integrable martingale then it converges to a random variable Y almost surely and in L^1 . Conversely, if $M(t)$ is a martingale that converges in L^1 to a random variable Y , then $M(t)$ is uniformly integrable, and it converges almost surely to Y . In any case $M(t) = E(Y|\mathcal{F}_t)$.*

Example 7.3: (Exponential martingale of Brownian motion.)

Let $M(t) = e^{B(t)-t/2}$. Then $M(t)$, $t \geq 0$ is a martingale. Since it is positive, it converges by Corollary 7.12 almost surely to a limit Y . By the Law of Large Numbers for Brownian motion $B(t)/t$ converges almost surely to zero. Thus $M(t) = e^{t(B(t)/t-1/2)} \rightarrow 0$ as $t \rightarrow \infty$. Thus $Y = 0$ almost surely. Therefore $M(t)$ is not uniformly integrable, as $EY = 0 \neq 1 = EM(t)$.

Example 7.4: Let $f(s)$ be non-random, such that $\int_0^\infty f^2(s)ds < \infty$. We show that $M(t) = \int_0^t f(s)dB(s)$ is a uniformly integrable martingale and find a representation for the closing random variable.

Since $\int_0^\infty f^2(s)ds < \infty$, $\int_0^t f(s)dB(s)$ is defined for all $t > 0$ and is a martingale. Since $\sup_{t>0} E(M^2(t)) = \sup_{t>0} \int_0^t f^2(s)ds = \int_0^\infty f^2(s)ds < \infty$, $M(t)$ is uniformly integrable. Thus it converges almost surely to Y . Convergence is also in L^1 , that is $E|M(t) - Y| \rightarrow 0$ as $t \rightarrow \infty$. Denote $Y = M(\infty) = \int_0^\infty f(s)dB(s)$, then we have shown that $Y - M(t) = \int_t^\infty f(s)dB(s)$ converges to zero almost surely and in L^1 . Y is the closing variable. Indeed,

$$E(Y|\mathcal{F}_t) = E(M(\infty)|\mathcal{F}_t) = E(\int_0^\infty f(s)dB(s)|\mathcal{F}_t) = \int_0^t f(s)dB(s) = M(t).$$

Example 7.5: A bounded positive martingale $M(t) = E(I(Y > 0)|\mathcal{F}_t)$ with $Y = \int_0^\infty f(s)dB(s)$, where $f(s)$ is non-random and $\int_0^\infty f^2(s)ds < \infty$, from the previous example.

$$\begin{aligned} M(t) &= E(I(Y > 0)|\mathcal{F}_t) = P(Y > 0|\mathcal{F}_t) \\ &= P\left(\int_t^\infty f(s)dB(s) > -\int_0^t f(s)dB(s) \middle| \mathcal{F}_t\right) \\ &= \Phi\left(\frac{\int_0^t f(s)dB(s)}{\sqrt{\int_t^\infty f^2(s)ds}}\right), \end{aligned} \tag{7.6}$$

where the last equality is due to normality of the Itô integral for a non-random f .

By taking f to be zero on (T, ∞) , a result is obtained for martingales of the form $E(I(\int_0^T f(s)dB(s) > 0)|\mathcal{F}_t)$. In particular, by taking $f(s) = 1_{[0,T]}(s)$, we obtain that

$$\Phi(B(t)/\sqrt{T-t})$$

is a positive bounded martingale on $[0, T]$. Its distribution for $t \leq T$ is left as an exercise.

7.4 Optional Stopping

In this section we consider results on stopping martingales at random times. Recall that a random time τ is called a stopping time if for any $t > 0$ the sets $\{\tau \leq t\} \in \mathcal{F}_t$. For filtrations generated by a process X , τ is a stopping time if it is possible to decide whether τ has occurred or not by observing the process up to time t . A martingale stopped at a random time τ is the process $M(t \wedge \tau)$. A Basic Stopping result, given here without proof, states that a martingale stopped at a stopping time is a martingale, in particular $EM(\tau \wedge t) = EM(0)$. This equation is used most frequently.

Theorem 7.14 *If $M(t)$ is a martingale and τ is a stopping time, then the stopped process $M(\tau \wedge t)$ is a martingale. Moreover,*

$$EM(\tau \wedge t) = EM(0). \quad (7.7)$$

This result was proved in discrete time (see Theorem 3.39). We refer to (7.7) as the Basic Stopping equation.

Remark 7.1: We stress that in this theorem $M(\tau \wedge t)$ is a martingale with respect to the original filtration $\mathcal{F}_t$. Since it is adapted to $\mathcal{F}_{\tau \wedge t}$, it is also an $\mathcal{F}_{\tau \wedge t}$ -martingale (see Exercise 7.1).

Example 7.6: (Exit of Brownian Motion from an Interval)

Let $B(t)$ be Brownian motion started at x and τ be the first time when $B(t)$ exits the interval (a, b) , $a < x < b$, that is, $\tau = \inf\{t : B(t) = a \text{ or } b\}$. Clearly, τ is a stopping time. By the basic stopping result (7.7), $EB(t \wedge \tau) = B(0) = x$. By definition of τ , $|B(t \wedge \tau)| \leq \max(|a|, |b|)$. Thus one can take $t \rightarrow \infty$, and use dominated convergence to obtain $EB(\tau) = x$. But $B(\tau) = b$ with probability p and $B(\tau) = a$ with probability $1 - p$. From these equations we obtain that $p = (x - a)/(b - a)$ is the probability that Brownian motion reaches b before it reaches a .

If M is a martingale then $EM(t) = EM(0)$. If τ is a stopping time, then $EM(\tau)$ may be different to $EM(0)$, as the next example shows.

Example 7.7: Let $B(t)$ be Brownian motion started at 0 and τ is the hitting time of 1. Then by definition $B(\tau) = 1$ and $EB(\tau) = 1 \neq 0 = EB(0)$.

However, under some additional assumptions on the martingale or on the stopping time, the random stopping does not alter the expected value. The following result gives sufficient conditions for optional stopping to hold.

Theorem 7.15 (Optional Stopping) *Let $M(t)$ be a martingale.*

1. *If $\tau \leq K < \infty$ is a bounded stopping time then $EM(\tau) = EM(0)$.*
2. *If $M(t)$ is uniformly integrable, then for any stopping time τ , $EM(\tau) = EM(0)$.*

The first statement follows from the Basic Stopping Result by taking $t > K$ and applying (7.7), $EM(0) = EM(t \wedge \tau) = EM(\tau)$.

Applied to gambling this shows that when betting on a martingale, on average no loss or gain is made, even if a clever stopping rule is used, provided it is bounded.

We don't give a proof of the second statement, the difficult point is in showing that $M(\tau)$ is integrable.

Theorem 7.16 *Let $M(t)$ be a martingale and τ a finite stopping time. If $E|M(\tau)| < \infty$, and*

$$\lim_{t \rightarrow \infty} E(M(t)I(\tau > t)) = 0, \quad (7.8)$$

then $EM(\tau) = EM(0)$.

PROOF: Write $M(\tau \wedge t)$ as

$$M(\tau \wedge t) = M(t)I(t < \tau) + M(\tau)I(t \geq \tau). \quad (7.9)$$

Using Basic Stopping Result (7.7), $EM(\tau \wedge t) = EM(0)$. Taking expectations in (7.9), we have

$$EM(0) = E(M(t)I(t < \tau)) + E(M(\tau)I(t \geq \tau)). \quad (7.10)$$

Now take the limit in (7.10) as $t \rightarrow \infty$. Since τ is finite, $I(t \geq \tau) \rightarrow I(\tau < \infty) = 1$. $|M(\tau)I(t \geq \tau)| \leq |M(\tau)|$, integrable. Hence $E(M(\tau)I(t \geq \tau)) \rightarrow EM(\tau)$ by dominated convergence. It is assumed that $E(M(t)I(t < \tau)) \rightarrow 0$ as $t \rightarrow \infty$, and the result follows. $\square$

The Basic Stopping result or Optional Stopping are used to find the distribution of stopping times for Brownian motion and Random Walks.

Example 7.8: (Hitting times of Brownian Motion)

We derive the Laplace transform of hitting times, from which it also follows that they are finite. Let $B(t)$ be a Brownian motion starting at 0, and $T_b = \inf\{t : B(t) = b\}$, $b > 0$. Consider the exponential martingale of Brownian motion $e^{uB(t) - u^2t/2}$, $u > 0$, stopped at T_b , $e^{uB(t \wedge T_b) - (t \wedge T_b)u^2/2}$. Using the Basic Stopping result (7.7)

$$Ee^{uB(t \wedge T_b) - (t \wedge T_b)u^2/2} = 1.$$

The martingale is bounded from above by e^{ub} and it is positive. If we take it as already proven that T_b is finite, $P(T_b < \infty) = 1$, then we obtain by taking $t \rightarrow \infty$ that $Ee^{ub - (t \wedge T_b)u^2/2} = 1$. Replacing u by $\sqrt{2u}$, we obtain the Laplace transform of T_b

$$\varphi_{T_b}(u) = E(e^{-uT_b}) = e^{-b\sqrt{2u}}. \quad (7.11)$$

We now show the finiteness of T_b . Write the expectation of the stopped martingale

$$\mathbb{E} \left(e^{ub - T_b u^2/2} I(T_b \leq t) \right) + \mathbb{E} \left(e^{uB(t) - tu^2/2} I(T_b > t) \right) = 1. \quad (7.12)$$

The term $\mathbb{E} \left(e^{uB(t) - tu^2/2} I(T_b > t) \right) \leq \mathbb{E} \left(e^{ub - tu^2/2} I(T_b > t) \right) \leq e^{ub - tu^2/2} \rightarrow 0$, as $t \rightarrow \infty$. Thus taking limits in the above equation (7.12),

$$\mathbb{E} \left(e^{ub - T_b u^2/2} I(T_b \leq t) \right) \rightarrow \mathbb{E} \left(e^{ub - T_b u^2/2} I(T_b < \infty) \right) = 1. \text{ Therefore}$$

$$\mathbb{E} \left(e^{-T_b u^2/2} I(T_b < \infty) \right) = e^{-ub}. \text{ But } e^{-T_b u^2} I(T_b = \infty) = 0, \text{ therefore by adding}$$

this term, we can write $\mathbb{E} \left(e^{-T_b u^2/2} \right) = e^{-ub}$. It follows in particular that

$P(T_b < \infty) = \lim_{u \downarrow 0} \psi(u) = 1$, and T_b is finite. Hence (7.11) is proved. The distribution of T_b corresponding to the transform (7.11) is given in Theorem 3.18.

The following result is in some sense the converse to the Optional Stopping Theorem.

Theorem 7.17 *Let $X(t)$, $t \geq 0$, be such that for any bounded stopping time τ , $X(\tau)$ is integrable and $\mathbb{E}X(\tau) = \mathbb{E}X(0)$. Then $X(t)$, $t \geq 0$ is a martingale.*

PROOF: The proof consists of checking the martingale property by using appropriate stopping times. Since a deterministic time t is a stopping time, $X(t)$ is integrable. Without loss of generality take $X(0) = 0$. Next we show that for $t > s$, $\mathbb{E}(X(t)|\mathcal{F}_s) = X(s)$. In other words, we need to show that for any $s < t$ and any set $B \in \mathcal{F}_s$

$$\mathbb{E}(X(t)I(B)) = \mathbb{E}(X(s)I(B)). \quad (7.13)$$

Fix a set $B \in \mathcal{F}_s$ and for any $t > s$, define a stopping time $\tau = sI(B) + tI(B^c)$. We have $\mathbb{E}(X(\tau)) = \mathbb{E}(X(s)I(B)) + \mathbb{E}(X(t)I(B^c))$. Since $\mathbb{E}X(\tau) = 0$, $\mathbb{E}(X(s)I(B)) = \mathbb{E}X(\tau) - \mathbb{E}(X(t)I(B^c)) = -\mathbb{E}(X(t)I(B^c))$.

As the right hand side of the above equality does not depend on s , it follows that (7.13) holds. □

The following result is sometimes known as the Optional Sampling Theorem (see for example, Rogers and Williams (1990)).

Theorem 7.18 (Optional Sampling) *Let $M(t)$ be a uniformly integrable martingale, and $\tau_1 \leq \tau_2 \leq \infty$ two stopping times. Then*

$$\mathbb{E}(M(\tau_2)|\mathcal{F}_{\tau_1}) = M(\tau_1), \text{ a.s.} \quad (7.14)$$

Optional Stopping of Discrete Time Martingales

We consider next the case of discrete time $t = 0, 1, 2, \dots$, and martingales arising in a Random Walk.

Gambler's Ruin

Consider a game played by two people by betting on the outcomes of tosses of a coin. You win \$1 if Heads come up and lose \$1 if Tails come up. The game stops when one party has no money left. You start with x , and your opponent with b dollars. Then S_n , the amount of money you have at time n is a Random Walk (see Section 3.12). The Gambler's ruin problem is to find the probabilities of ruin of the players.

In this game the loss of one person is the gain of the other (a zero sum game). Assuming that the game will end in a finite time τ (this fact will be shown later), it follows that the ruin probabilities of the players add up to one.

Consider first the case of the fair coin. Then

$$S_n = x + \sum_{i=1}^n \xi_i, \quad P(\xi_i = 1) = \frac{1}{2}, \quad P(\xi_i = -1) = \frac{1}{2},$$

is a martingale (see Theorem 3.33). Let τ be the time when the game stops, the first time the amount of money you have is equal to 0 (your ruin) or $x + b$ (your opponent's ruin). Then τ is a stopping time. Denote by u the probability of your ruin. It is the probability of you losing your initial capital x before winning b dollars. Thus

$$P(S_\tau = 0) = u \quad \text{and} \quad P(S_\tau = x + b) = 1 - u. \quad (7.15)$$

Formally applying the Optional Stopping Theorem

$$E(S_\tau) = S_0 = x. \quad (7.16)$$

But

$$E(S_\tau) = (x + b) \times (1 - u) + 0 \times u = (x + b)u.$$

These equations give

$$u = \frac{b}{x + b}. \quad (7.17)$$

So that the ruin probabilities are given by a simple calculation using martingale stopping.

We now justify the steps. S_n is a martingale, and τ is a stopping time. By Theorem 7.14 the stopped process $S_{n \wedge \tau}$ is a martingale. It is non-negative and bounded by $x + b$, by the definition of τ . Thus $S_{n \wedge \tau}$ is a uniformly integrable martingale. Hence it converges almost surely to a finite limit Y , (with $EY = x$), $\lim_{n \rightarrow \infty} S_{n \wedge \tau} = Y$. By Theorem 3.33 $S_n^2 - n$ is a martingale, and so is $S_{n \wedge \tau}^2 - n \wedge \tau$. Thus for all n by taking expectation

$$E(S_{n \wedge \tau}^2) = E(n \wedge \tau) + E(S_0^2). \quad (7.18)$$

By dominated convergence, the lhs has a finite limit, therefore there is a finite limit $\lim_{n \rightarrow \infty} E(n \wedge \tau)$. Expanding this, $E(n \wedge \tau) \geq nP(\tau > n)$, we can see that for a limit to exist it must be

$$\lim_{n \rightarrow \infty} P(\tau > n) = 0, \quad (7.19)$$

so that $P(\tau < \infty) = 1$, and τ is finite. (Note that a standard proof of finiteness of τ is done by using Markov Chain Theory, the property of recurrence of states in a Random Walk.) Writing $E(S_{n \wedge \tau}) = x$ and taking limits as $n \rightarrow \infty$, the equation (7.16) is obtained (alternatively, the conditions of the Optional Stopping Theorem hold). This concludes a rigorous derivation of the ruin probability in an unbiased Random Walk.

We now consider the case when the Random Walk is biased, $p \neq q$.

$$S_n = x + \sum_{i=1}^n \xi_i, \quad P(\xi_i = 1) = p, \quad P(\xi_i = -1) = q = 1 - p.$$

In this case the exponential martingale of the Random Walk $M_n = (q/p)^{S_n}$ is used (see Theorem 3.33). Stopping this martingale, we obtain the ruin probability

$$u = \frac{(q/p)^{b+x} - (q/p)^x}{(q/p)^{b+x} - 1}. \quad (7.20)$$

Justification of the equation $E(M_\tau) = M_0$ is similar to the previous case.

Hitting Times in Random Walks

Let S_n denote a Random Walk on the integers started at $S_0 = x$, $S_n = S_0 + \sum_{i=1}^n \xi_i$, $P(\xi_i = 1) = p$, $P(\xi_i = -1) = q = 1 - p$, with arbitrary p , and T_b the first hitting time of b , $T_b = \inf\{n : S_n = b\}$ (infimum of an empty set is infinity). Without loss of generality take the starting state $x = 0$, otherwise consider the process $S_n - x$. Consider hitting the level $b > 0$, for $b < 0$ consider the process $-S_n$.

We find the Laplace transform of T_b , $\psi(\lambda) = E(e^{-\lambda T_b})$, $\lambda > 0$, by stopping the exponential martingale of the Random Walk $M_n = e^{u S_n - n h(u)}$, where $h(u) = \ln E(e^{u \xi_1})$, and u is arbitrary (see Section 3.12).

$$E(M_{n \wedge T_b}) = E\left(e^{u S_{n \wedge T_b} - (n \wedge T_b) h(u)}\right) = 1. \quad (7.21)$$

Take u , so that $h(u) = \lambda > 0$. Write the expectation in (7.21) as

$$E\left(e^{u S_{n \wedge T_b} - (n \wedge T_b) h(u)}\right) = E\left(e^{u S_{T_b} - T_b h(u)} I(T_b \leq n)\right) + E\left(e^{u S_n - n h(u)} I(T_b > n)\right).$$

The first term equals to $E(e^{ub-T_b h(u)} I(T_b \leq n))$. The second term converges to zero, because by definition of T_b ,

$$E(e^{uS_n - nh(u)} I(T_b > n)) \leq E(e^{ub - nh(u)} I(T_b > n)) \leq e^{ub - nh(u)} \rightarrow 0.$$

Now taking limits in (7.21), using dominated convergence we obtain

$$E(e^{-h(u)T_b} I(T_b < \infty)) = e^{-ub}.$$

Note that $e^{-h(u)T_b} I(T_b = \infty) = 0$, therefore by adding this term, we can write

$$E(e^{-h(u)T_b}) = e^{-ub}. \quad (7.22)$$

This is practically the Laplace transform of T_b , it remains to replace $h(u)$ by λ , by taking $u = h^{(-1)}(\lambda)$, with $h^{(-1)}$ being the inverse of h . Thus the Laplace transform of T_b is given by

$$\psi(\lambda) = E(e^{-\lambda T_b}) = e^{-h^{(-1)}(\lambda)b}. \quad (7.23)$$

To find $h^{(-1)}(\lambda)$, solve $h(u) = \lambda$, which is equivalent to $E(e^{u\xi_1}) = e^\lambda$, or $pe^u + (1-p)e^{-u} = e^\lambda$. There are two values for $e^u = (e^\lambda \pm \sqrt{e^{2\lambda} - 4p(1-p)})/(2p)$, but only one corresponds to a Laplace transform, (7.22). Thus we have

$$\psi(\lambda) = E(e^{-\lambda T_b}) = \left(\frac{2p}{e^\lambda + \sqrt{e^{2\lambda} - 4p(1-p)}} \right)^b. \quad (7.24)$$

Using a general result on Laplace transform of a random variable,

$$P(T_b < \infty) = \lim_{\lambda \downarrow 0} \psi(\lambda) = \left(\frac{2p}{1 + |1 - 2p|} \right)^b. \quad (7.25)$$

It now follows that the hitting time T_b of b is finite if and only if $p \geq 1/2$. For $p < 1/2$, there is a positive probability that level b is never reached, $P(T_b = \infty) = 1 - (\frac{p}{1-p})^b$.

When the hitting time of level b is finite, it may or may not have a finite expectation. If $p \geq 1/2$ we have

$$E(T_b) = -\psi'(0) = \begin{cases} \frac{1}{2p-1} & \text{if } p > 1/2 \\ \infty & \text{if } p = 1/2. \end{cases}$$

Thus we have shown that when $p \geq 1/2$ any positive state will be reached from 0 in a finite time, but when $p = 1/2$ the average time for it to happen is infinite.

The results obtained above are known as transience ($p \neq 1/2$) and recurrence ($p = 1/2$) of the Random Walk, and are usually obtained by Markov Chains Theory.

Example 7.9: (Optional stopping of discrete time martingales)

Let $M(t)$ be a discrete time martingale and τ be a stopping time such that $E|M(\tau)| < \infty$.

1. If $E\tau < \infty$ and $|M(t+1) - M(t)| \leq K$, then $EM(\tau) = EM(0)$.
2. If $E\tau < \infty$ and $E(|M(t+1) - M(t)| | \mathcal{F}_t) \leq K$, then $EM(\tau) = EM(0)$.

PROOF: We prove the first statement.

$M(t) = M(0) + \sum_{i=0}^{t-1} (M(i+1) - M(i))$. This together with the bound on increments gives

$$M(t) \leq |M(0)| + \sum_{i=0}^{t-1} |M(i+1) - M(i)| \leq |M(0)| + Kt.$$

Take for simplicity non-random $M(0)$ Then

$$EM(t)I(\tau > t) \leq |M(0)|P(\tau > t) + KtP(\tau > t).$$

The last term converges to zero, $tP(\tau > t) \leq E(\tau I(\tau > t)) \rightarrow 0$, by dominated convergence due to $E(\tau) < \infty$. Thus condition (7.8) holds, and the result follows. The proof of the second statement is similar and is left as an exercise. $\square$

7.5 Localization and Local Martingales

As it was seen earlier in Chapter 4, Itô integrals $\int_0^t X(s)dB(s)$ are martingales under the additional condition $\int_0^t X^2(s)ds < \infty$. In general, stochastic integrals with respect to martingales are only local martingales rather than true martingales. This is the main reason for introducing local martingales. We have also seen that for the calculation of expectations stopping and truncations are often used. These ideas give rise to the following

Definition 7.19 *A property of a stochastic process $X(t)$ is said to hold locally if there exists a sequence of stopping times τ_n , called the localizing sequence, such that $\tau_n \uparrow \infty$ as $n \rightarrow \infty$ and for each n the stopped processes $X(t \wedge \tau_n)$ has this property.*

For example, the uniform integrability property holds locally for any martingale. By Theorem 7.13 a martingale convergent in L^1 is uniformly integrable. Here $M(t \wedge n) = M(n)$ for $t > n$, and therefore $\tau_n = n$ is a localizing sequence.

Local martingales are defined by localizing the martingale property.

Definition 7.20 *An adapted process $M(t)$ is called a local martingale if there exists a sequence of stopping times τ_n , such that $\tau_n \uparrow \infty$ and for each n the stopped processes $M(t \wedge \tau_n)$ is a uniformly integrable martingale in t .*

As we have just seen, any martingale is a local martingale. Examples of local martingales which are not martingales are given below.

Example 7.10: $M(t) = 1/|B(t)|$, where $B(t)$ is the three-dimensional Brownian motion, $B(0) = \mathbf{x} \neq \mathbf{0}$. We have seen in Example 6.17 that if D_r is the complementary set to the ball of radius r centered at the origin, $D_r = \{\mathbf{z} : |\mathbf{z}| > r\}$, then $f(\mathbf{z}) = 1/|\mathbf{z}|$ is a harmonic function for the Laplacian on D_r . Consequently $1/|B(t \wedge \tau_{D_r})|$ is a martingale, where τ_{D_r} is the time of exit from D_r . Take now τ_n be the exit time from $D_{1/n}$, that is, $\tau_n = \inf\{t > 0 : |B(t)| = 1/n\}$. Then for any fixed n , $1/|B(t \wedge \tau_n)|$ is a martingale. τ_n increase to, say, τ and by continuity, $B(\tau) = \mathbf{0}$. As Brownian motion in three dimensions never visits the origin (see Example 6.14), it follows by continuity that τ is infinite. Thus $M(t)$ is a local martingale. To see that it is not a true martingale, recall that in three dimensions Brownian motion is transient and $|B(t)| \rightarrow \infty$ as $t \rightarrow \infty$. Therefore $EM(t) \rightarrow 0$, whereas $EM(0) = 1/|\mathbf{x}|$. Since the expectation of a martingale is constant, $M(t)$ is not a martingale.

Example 7.11: (Itô integrals.)

Let $M(t) = \int_0^t e^{B^2(s)} dB(s)$, $t > 1/4$, where B is Brownian motion in one dimension with $B(0) = 0$. Let $\tau_n = \inf\{t > 0 : e^{B^2(t)} = n\}$. Then for $t \leq \tau_n$, the integrand is bounded by n . By the martingale property of Itô integrals, $M(t \wedge \tau_n)$ is a martingale in t for any n . By continuity, $\exp(B^2(\tau)) = \infty$, thus $\tau_n \rightarrow \tau = \infty$. Therefore $M(t)$ is a local martingale. To see that it is not a martingale notice that for $t > 1/4$, $E(e^{B^2(t)}) = \infty$, implying that $M(t)$ is not integrable.

Remark 7.2: Note that it is not enough for a local martingale to be integrable in order to be a true martingale. For example, positive local martingales are integrable, but in general they are not martingales, but only supermartingales (see Theorem 7.23) below. Even uniformly integrable local martingales may not be martingales. However, if a local martingale is dominated by an integrable random variable then it is a martingale.

Theorem 7.21 *Let $M(t)$, $0 \leq t < \infty$, be a local martingale such that $|M(t)| \leq Y$, with $EY < \infty$. Then M is a uniformly integrable martingale.*

PROOF: Let τ_n be a localizing sequence. Then for any n and $s < t$

$$E(M(t \wedge \tau_n) | \mathcal{F}_s) = M(s \wedge \tau_n). \quad (7.26)$$

M is clearly integrable, since $E|M(t)| \leq EY < \infty$. Since $\lim_{n \rightarrow \infty} M(t \wedge \tau_n) = M(t)$, by dominated convergence of conditional expectations $\lim_{n \rightarrow \infty} E(M(t \wedge \tau_n) | \mathcal{F}_s) = E(M(t) | \mathcal{F}_s)$. Since $\lim_{n \rightarrow \infty} M(s \wedge \tau_n) = M(s)$, the martingale property is established by taking limits in (7.26). If a martingale is dominated by an integrable random variable then it is uniformly integrable (see Theorem 7.6).

□

Corollary 7.22 *Let $M(t)$, $0 \leq t < \infty$, be a local martingale such that for all t , $E(\sup_{s \leq t} |M(s)|) < \infty$. Then it is a martingale, and as such it is uniformly*

integrable on any finite interval $[0, T]$. If in addition $E(\sup_{t \geq 0} |M(t)|) < \infty$, then $M(t)$, $t \geq 0$, is uniformly integrable on $[0, \infty)$.

In financial applications we meet positive local martingales.

Theorem 7.23 *A non-negative local martingale $M(t)$, $0 \leq t \leq T$, is a supermartingale, that is, $EM(t) < \infty$, and for any $s < t$, $E(M(t)|\mathcal{F}_s) \leq M(s)$.*

PROOF: Let τ_n be a localizing sequence. Then since $M(t \wedge \tau_n) \geq 0$, by Fatou's lemma

$$E\left(\liminf_{n \rightarrow \infty} M(t \wedge \tau_n)\right) \leq \liminf_{n \rightarrow \infty} E(M(t \wedge \tau_n)). \quad (7.27)$$

Since the limit exists, the lower limit is the same, that is, $\lim_{n \rightarrow \infty} M(t \wedge \tau_n) = M(t)$ implies $\liminf_{n \rightarrow \infty} M(t \wedge \tau_n) = M(t)$. But $EM(t \wedge \tau_n) = EM(0 \wedge \tau_n) = EM(0)$ by the martingale property of $M(t \wedge \tau_n)$. Therefore by taking limits, $EM(t) \leq EM(0)$, so that M is integrable. The supermartingale property is established similarly. Using Fatou's lemma for conditional expectations,

$$E\left(\liminf_{n \rightarrow \infty} M(t \wedge \tau_n) | \mathcal{F}_s\right) \leq \liminf_{n \rightarrow \infty} E(M(t \wedge \tau_n) | \mathcal{F}_s) = M(s \wedge \tau_n), \quad (7.28)$$

Taking limits as $n \rightarrow \infty$ we obtain $E(M(t)|\mathcal{F}_s) \leq M(s)$ almost surely. □

From this result and Theorem 7.3 we obtain

Theorem 7.24 *A non-negative local martingale $M(t)$, $0 \leq t \leq T$, is a martingale if and only if $EM(T) = M(0)$.*

For a general local martingale a necessary and sufficient condition to be a uniformly integrable martingale is described in terms of the property of Dirichlet class (D). This class of processes also arises in other areas of calculus and is given in the next section.

Dirichlet Class (D)

Definition 7.25 *A process X is of Dirichlet class, (D), if the family $\{X(\tau) : \tau \text{ a finite stopping time}\}$ is uniformly integrable.*

Any uniformly integrable martingale M is of class (D). Indeed, by Theorem 7.13, M is closed by $Y = M(\infty)$, $M(\tau) = E(Y|\mathcal{F}_\tau)$, and the last family is uniformly integrable.

Using localization one can show the other direction, and we have a theorem

Theorem 7.26 *A local martingale M is a uniformly integrable martingale if and only if it is of class (D).*

PROOF: Suppose that M is a local martingale of class (D) .

Let τ_n be a localizing sequence, so that $M(t \wedge \tau_n)$ is a uniformly integrable martingale in t . Then for $s < t$,

$$M(s \wedge \tau_n) = E(M(t \wedge \tau_n) | \mathcal{F}_s). \quad (7.29)$$

The martingale property of M is obtained by taking $n \rightarrow \infty$ in both sides of the equation.

Since $\tau_n \rightarrow \infty$, $M(s \wedge \tau_n) \rightarrow M(s)$ almost surely. $s \wedge \tau_n$ is a finite stopping time, and because M is in (D) , the sequence of random variables $\{M(s \wedge \tau_n)\}_n$ is uniformly integrable. Thus $M(s \wedge \tau_n) \rightarrow M(s)$ also in L^1 , that is,

$$E|M(s \wedge \tau_n) - M(s)| \rightarrow 0. \quad (7.30)$$

Using the properties of conditional expectation,

$$\begin{aligned} E|E(M(t \wedge \tau_n) | \mathcal{F}_s) - E(M(t) | \mathcal{F}_s)| &= E|E(M(t \wedge \tau_n) - M(t) | \mathcal{F}_s)| \\ &\leq E(E|M(t \wedge \tau_n) - M(t)| | \mathcal{F}_s) \\ &= E|M(t \wedge \tau_n) - M(t)|. \end{aligned}$$

The latter converges to zero by (7.30). This implies $E(M(t \wedge \tau_n) | \mathcal{F}_s) \rightarrow E(M(t) | \mathcal{F}_s)$ as $n \rightarrow \infty$. Taking limits in (7.29) as $n \rightarrow \infty$ establishes the martingale property of M . Since it is in (D) , by taking $\tau = t$, it is uniformly integrable. □

7.6 Quadratic Variation of Martingales

Quadratic variation of a process $X(t)$ is defined as a limit in probability

$$[X, X](t) = \lim \sum_{i=1}^n (X(t_i^n) - X(t_{i-1}^n))^2, \quad (7.31)$$

where the limit is taken over partitions:

$$0 = t_0^n < t_1^n < \dots < t_n^n = t,$$

with $\delta_n = \max_{0 \leq i \leq n} (t_i^n - t_{i-1}^n) \rightarrow 0$. If $M(t)$ is a martingale, then $M^2(t)$ is a submartingale, and its mean increases (unless $M(t)$ is a constant). By compensating $M^2(t)$ by some increasing process, it is possible to make it into a martingale. The process which compensates $M^2(t)$ to a martingale turns out to be the quadratic variation process of M . It can be shown that quadratic variation of martingales exists and is characterized by the above property.

Theorem 7.27

1. Let $M(t)$ be a martingale with finite second moments, $E(M^2(t)) < \infty$ for all t . Then its quadratic variation process $[M, M](t)$ defined in (7.31) exists, moreover $M^2(t) - [M, M](t)$ is a martingale.
2. If M is a local martingale, then $[M, M](t)$ exists, moreover $M^2(t) - [M, M](t)$ is a local martingale.

PROOF: We outline the proof of the first statement only. The second follows for locally square integrable martingales by localization. For local martingales the result follows from representation of quadratic variation by means of stochastic integrals. For a full proof see for example, Liptser and Shiryaev (1989), p.56-59.

$$E(M(t)M(s)) = EE(M(t)M(s)|\mathcal{F}_s) = E(M(s)E(M(t)|\mathcal{F}_s)) = E(M^2(s)). \quad (7.32)$$

Using this it is easy to obtain

$$E(M(t) - M(s))^2 = E(M^2(t)) - E(M^2(s)). \quad (7.33)$$

It is easy to see that the sums in the definition of quadratic variation $[M, M](t)$ have constant mean, that of $EM^2(t)$. It is possible, but is not easy, to prove that these sums converge in probability to the limit $[M, M](t)$. Now using property (7.33), we can write

$$\begin{aligned} E(M^2(t) - M^2(s)|\mathcal{F}_s) &= E((M(t) - M(s))^2|\mathcal{F}_s) \\ &= E\left(\sum_{i=0}^{n-1} (M(t_{i+1}) - M(t_i))^2|\mathcal{F}_s\right), \end{aligned} \quad (7.34)$$

where $\{t_i\}$ is a partition of $[s, t]$. Taking the limit as the size of the partition goes to zero, we obtain

$$E(M^2(t) - M^2(s)|\mathcal{F}_s) = E([M, M](t) - [M, M](s)|\mathcal{F}_s). \quad (7.35)$$

Rearranging, we obtain the martingale property of $M^2(t) - [M, M](t)$. □

For the next result note that if M is a martingale, then for any t

$$E(M(t) - M(0))^2 = E(M^2(t)) - E(M^2(0)),$$

which shows that $E(M^2(t)) > E(M^2(0))$ unless $M(t) = M(0)$ a.s. Thus M^2 can not be a martingale on $[0, t]$, unless $M(t) = M(0)$. If $M(t) = M(0)$, then for all $s < t$, $M(s) = E(M(t)|\mathcal{F}_s) = M(0)$, and M is a constant on $[0, t]$.

Theorem 7.28 *Let M be a martingale with $M(0) = 0$. If for some t , $M(t)$ is not identically zero, then $[M, M](t) > 0$. Conversely, if $[M, M](t) = 0$, then $M(s) = 0$ a.s. for all $s \leq t$. The result also holds for local martingales.*

PROOF: We prove the result for square integrable martingales; for local martingales it can be shown by localization. Suppose that $[M, M](t) = 0$ for some $t > 0$. Then, since $[M, M]$ is non-decreasing, $[M, M](s) = 0$ for all $s \leq t$. By Theorem 7.27 $M^2(s)$, $s \leq t$, is a martingale. In particular, $E(M^2(t)) = 0$. This implies that $M(t) = 0$ a.s., which is a contradiction. Therefore $[M, M](t) > 0$.

Conversely, if $[M, M](t) = 0$, the same argument shows that $M(t) = 0$ a.s., and by the martingale property $M(s) = 0$ for all $s \leq t$. □

It also follows from the proof that M and $[M, M]$ have same intervals of constancy. This theorem implies remarkably that a continuous martingale which is not a constant has infinite variation on any interval.

Theorem 7.29 *Let M be a continuous local martingale, and fix any t . If $M(t)$ is not identically equal to $M(0)$, then M has infinite variation over $[0, t]$.*

PROOF: $M(t) - M(0)$ is a martingale, null at zero, with its value at time t not equal identically to zero. By the above theorem M has a positive quadratic variation on $[0, t]$, $[M, M](t) > 0$. By Theorem 1.10 a continuous process of finite variation on $[0, t]$ has zero quadratic variation over this interval. Therefore M must have infinite variation over $[0, t]$. □

Corollary 7.30 *If a continuous local martingale has finite variation over an interval, then it must be a constant over that interval.*

Remark 7.3: Note that there are martingales with finite variation, but by the previous result they can not be continuous. An example of such a martingale is the Poisson process martingale $N(t) - t$.

7.7 Martingale Inequalities

$M(t)$ denotes a martingale or a local martingale on the interval $[0, T]$ with possibly $T = \infty$.

Theorem 7.31 *If $M(t)$ is a martingale (or a positive submartingale) then for $p \geq 1$*

$$P(\sup_{s \leq t} |M(s)| \geq a) \leq a^{-p} \sup_{s \leq t} E(|M(s)|^p). \quad (7.36)$$

If $p > 1$, then

$$\mathbf{E}\left(\sup_{s \leq t} |M(s)|^p\right) \leq \left(\frac{p}{p-1}\right)^p \mathbf{E}(|M(t)|^p). \quad (7.37)$$

The case of $p = 2$ is called Doob's inequality for martingales.

$$\mathbf{E}\left(\sup_{s \leq T} M(s)^2\right) \leq 4\mathbf{E}(M^2(T)). \quad (7.38)$$

As a consequence, if for $p > 1$, $\sup_{t \leq T} \mathbf{E}(|M(t)|^p) < \infty$, then $M(t)$ is uniformly integrable (This is a particular case of Theorem 7.7).

Theorem 7.32 *If M is locally square integrable martingale with $M(0) = 0$, then*

$$\mathbf{P}(\sup_{t \leq T} |M(t)| > a) \leq a^{-2} \mathbf{E}([M, M](T)). \quad (7.39)$$

Theorem 7.33 (Davis' Inequality) *There are constants $c > 0$ and $C < \infty$ such that for any local martingale $M(t)$, null at zero,*

$$c\mathbf{E}\left(\sqrt{[M, M](T)}\right) \leq \mathbf{E}\left(\sup_{t \leq T} |M(t)|\right) \leq C\mathbf{E}\left(\sqrt{[M, M](T)}\right). \quad (7.40)$$

Theorem 7.34 (Burkholder-Gundy Inequality) *There are constants c_p and C_p depending only on p , such that for any local martingale $M(t)$, null at zero,*

$$c_p \mathbf{E}\left([M, M](T)^{p/2}\right) \leq \mathbf{E}\left(\left(\sup_{t \leq T} |M(t)|\right)^p\right) \leq C_p \mathbf{E}\left([M, M](T)^{p/2}\right), \quad (7.41)$$

for $1 < p < \infty$. If moreover, $M(t)$ is continuous, then the result holds also for $0 < p \leq 1$.

The above inequalities hold when T is a stopping time.

Proofs of these inequalities involve concepts of stochastic calculus for general processes and can be found, for example, in Protter (1992), Rogers and Williams (1990), Liptser and Shiriyayev (1989).

We use the above inequalities to give sufficient conditions for a local martingale to be a true martingale.

Theorem 7.35 *Let $M(t)$ be a local martingale, null at zero, such that $\mathbf{E}\left(\sqrt{[M, M](t)}\right) < \infty$ for all t . Then $M(t)$ is a uniformly integrable martingale on $[0, T]$, for any finite T .*

If moreover, $\mathbf{E}[M, M](t) < \infty$, then $M(t)$ is a martingale with $\mathbf{E}M^2(t) = \mathbf{E}[M, M](t) < \infty$ for all t .

If $\sup_{t < \infty} \mathbf{E}[M, M](t) < \infty$, then $M(t)$ is a square integrable martingale.

PROOF: By the Davis inequality, $\sup_{t \leq T} |M(t)|$ is an integrable random variable, $E(\sup_{t \leq T} |M(t)|) \leq CE \left(\sqrt{[M, M](T)} \right) < \infty$. Thus $M(t)$ is dominated by an integrable random variable on any finite time interval. Therefore it is a uniformly martingale by Theorem 7.21, and the first claim is proved.

The condition $E[M, M](t) < \infty$ implies the previous condition $E \left(\sqrt{[M, M](t)} \right) < \infty$, as for $X \geq 0$, $E(X) \geq (E(\sqrt{X}))^2$, due to $Var(\sqrt{X}) \geq 0$. Thus M is a martingale. Alternatively, use the Burkholder-Gundy inequality with $p = 2$. Next, recall that by Theorem 7.27, if $M(t)$ is a martingale with $E(M^2(t)) < \infty$, then $M^2(t) - [M, M](t)$ is a martingale. In particular, for any finite t

$$E(M^2(t)) = E[M, M](t),$$

and the second statement is proved. To prove the third, notice that since both sides in the above equation are non-decreasing, they have a limit. Since by assumption $\lim_{t \rightarrow \infty} E[M, M](t) < \infty$, $\sup_{t < \infty} EM^2(t) < \infty$, and $M(t)$, $0 \leq t < \infty$ is a square integrable martingale. □

Application to Itô integrals

Let $X(t) = \int_0^t H(s)dB(s)$. Being an Itô integral, X is a local martingale. Its quadratic variation is given by $[X, X](t) = \int_0^t H^2(s)ds$. The Burkholder-Gundy inequality with $p = 2$ gives $E(\sup_{t \leq T} X^2(t)) \leq CE([X, X](T)) = E \int_0^T H^2(s)ds$. If $E \left(\int_0^T H^2(s)ds \right) < \infty$, then $X(t)$ is a square integrable martingale. Thus we recover the known fact that $E(X^2(t)) = E \left(\int_0^t H^2(s)ds \right)$.

The Davis inequality gives $E(\sup_{t \leq T} \int_0^t H(s)dB(s)) \leq CE \left(\sqrt{\int_0^T H^2(s)ds} \right)$. Thus the condition

$$E \left(\sqrt{\int_0^t H^2(s)ds} \right) < \infty \tag{7.42}$$

is a sufficient condition for the Itô integral to be a martingale and, in particular, to have zero mean. This condition, however, does not assure second moments.

7.8 Continuous Martingales. Change of Time

Brownian motion is the basic continuous martingale from which all continuous martingales can be constructed, either by random change of time, given in this section, or by stochastic integration, as will be seen in the next chapter. The starting point is a result that characterizes a Brownian motion.

Levy's Characterization of Brownian Motion

Theorem 7.36 (Levy) *A process M with $M(0) = 0$ is a Brownian motion if and only if it is a continuous local martingale with quadratic variation process $[M, M](t) = t$.*

PROOF: If M is a Brownian motion, then it is a continuous martingale with $[M, M](t) = t$.

Let $M(t)$ be a continuous local martingale with $[M, M](t) = t$. Then $uM(t)$ is a continuous local martingale with $[uM, uM](t) = u^2 t$. We show that

$$U(t) = e^{uM(t) - u^2 t/2} = e^{uM(t) - [uM, uM](t)/2} \quad (7.43)$$

is a martingale. Once this is established, the rest of the proof follows by an application of the martingale property.

The general theory of integration with respect to martingales is required to show the martingale property of $U(t)$. It is an easy corollary of a general result on stochastic exponential martingales (Theorem 8.17, Corollary 8.18). Writing the martingale property, we have

$$\mathbb{E}\left(e^{uM(t) - u^2 t/2} | \mathcal{F}_s\right) = e^{uM(s) - u^2 s/2}, \quad (7.44)$$

from which it follows that

$$\mathbb{E}\left(e^{u(M(t) - M(s))} | \mathcal{F}_s\right) = e^{u^2(t-s)/2}. \quad (7.45)$$

Since the right hand side of (7.45) is non-random, it follows that $M(t)$ has independent increments. Taking expectation in (7.45), we obtain

$$\mathbb{E}\left(e^{u(M(t) - M(s))}\right) = e^{u^2(t-s)/2}, \quad (7.46)$$

which shows that the increment of the martingale $M(t) - M(s)$ has Normal distribution with mean zero and variance $(t - s)$. Therefore M is a continuous process with independent Gaussian increments, hence it is Brownian motion. $\square$

Example 7.12: Any solution of Tanaka's SDE in Example 5.15 is a Brownian motion (weak uniqueness).

$dX(t) = \text{sign}(X(t))dB(t)$, where $\text{sign}(x) = 1$ if $x \geq 0$ and -1 if $x < 0$. $X(0) = 0$. $X(t) = \int_0^t \text{sign}(X(s))dB(s)$. Since it is an Itô integral, it is a local martingale (even a martingale, as the condition for it to be a martingale holds). It is continuous, and its quadratic variation is given by $[X, X](t) = \int_0^t \text{sign}^2(X(s))ds = t$. Therefore it is a Brownian motion.

Change of Time for Martingales

The main result below states that a continuous martingale M is a Brownian motion with a change of time, where time is measured by the quadratic variation $[M, M](t)$, namely, there is a Brownian motion $B(t)$, such that $M(t) = B([M, M](t))$. This $B(t)$ is constructed from $M(t)$. Define

$$\tau_t = \inf\{s : [M, M](s) > t\}. \quad (7.47)$$

If $[M, M](t)$ is strictly increasing, then τ_t is its inverse.

Theorem 7.37 (Dambis, Dubins-Schwarz) *Let $M(t)$ be a continuous martingale, null at zero, such that $[M, M](t)$ is non-decreasing to ∞ , and τ_t defined by (7.47). Then the process $B(t) = M(\tau_t)$ is a Brownian motion with respect to the filtration $\mathcal{F}_{\tau_t}$. Moreover, $[M, M](t)$ is a stopping time with respect to this filtration, and the martingale M can be obtained from the Brownian motion B by the change of time $M(t) = B([M, M](t))$. The result also holds when M is a continuous local martingale.*

We outline the idea of the proof, for details see, for example, Rogers and Williams (1990), p.64, Karatzas and Shreve (1988) p.174, Protter (1992) p.81, Revuz and Yor (1998) p. 181.

PROOF: Let $M(t)$ be a local martingale. τ_t defined by (7.47) are finite stopping times, since $[M, M](t) \rightarrow \infty$. Thus $\mathcal{F}_{\tau_t}$ are well defined, (see Chapter 2 for the definition of $\mathcal{F}_\tau$). Note that $\{[M, M](s) \leq t\} = \{\tau_t \geq s\}$. This implies that $[M, M](s)$ are stopping times for $\mathcal{F}_{\tau_t}$. Since $[M, M](s)$ is continuous $[M, M](\tau_t) = t$. Let $X(t) = M(\tau_t)$. Then it is a continuous local martingale, since M and $[M, M]$ have the same intervals of constancy (see the comment following Theorem 7.28). Using Theorem 7.27 we obtain $EX^2(t) = E[X, X](t) = E[M, M](\tau_t) = t$. Thus X is a Brownian motion by Levy's characterization Theorem 7.36. The second part is proven as follows. Recall that M and $[M, M]$ have the same intervals of constancy. Thus $X([M, M](t)) = M(\tau_{[M, M](t)}) = M(t)$.

□

Example 7.13: Let $M(t) = \int_0^t f(s)dB(s)$, with f continuous and non-random. Then M is a Gaussian martingale. Its quadratic variation is given by $[M, M](t) = \int_0^t f^2(s)ds$. For example, with $f(s) = s$, $M(t) = \int_0^t s dB(s)$ and $[M, M](t) = \int_0^t s^2 ds = t^3/3$. In this example $[M, M](t)$ is non-random and increasing. τ_t is given by its inverse, $\tau_t = (3t)^{1/3}$. Let $X(t) = M(\tau_t) = \int_0^{\sqrt[3]{3t}} s dB(s)$. Then, clearly, X is continuous, as a composition of continuous functions. It is also a martingale with quadratic variation $\tau_t^3/3 = t$. Hence, by the Levy's theorem, it is a Brownian motion, $X(t) = \hat{B}(t)$. By the above theorem, $M(t) = \hat{B}(t^3/3)$.

Example 7.14: If $M(t) = \int_0^t H(s)dB(s)$ is an Itô integral, then it is a local martingale with quadratic variation $[M, M](t) = \int_0^t H^2(s)ds$. If $\int_0^\infty H^2(s)ds = \infty$, then $M(t) = \hat{B}(\int_0^t H^2(s)ds)$, where $\hat{B}(t)$ is Brownian motion and can be recovered from $M(t)$ with the appropriate change of time.

Example 7.15: (Brownian Bridge as Time Changed Brownian motion)

The SDE for Brownian Bridge (5.34) contains as its only stochastic term $\int_0^t \frac{1}{T-s}dB(s)$. Since for any $t < T$, it is a continuous martingale with quadratic variation $[Y, Y](t) = \int_0^t \frac{1}{(T-s)^2}ds = \frac{t}{T(T-t)}$, it follows by the DDS Theorem

$$Y(t) = \hat{B}\left(\frac{t}{T(T-t)}\right),$$

for some Brownian motion $\hat{B}$. Therefore SDE (5.34) has the following representation

$$X(t) = a\left(1 - \frac{t}{T}\right) + b\frac{t}{T} + (T-t)\hat{B}\left(\frac{t}{T(T-t)}\right), \text{ for } 0 \leq t \leq T. \quad (7.48)$$

In this representation $t = T$ is allowed and understood by continuity, since the limit of $tB(1/t)$ as $t \rightarrow 0$ is zero by the Law of Large Numbers for Brownian motion.

Change of Time in SDEs

We use the Change of Time (DDS) Theorem for constructing weak solutions of some SDEs. Let

$$X(t) = \int_0^t \sqrt{f'(s)}dB(s), \quad (7.49)$$

where $f(t)$ is an adapted, positive, increasing, differentiable process, null at zero. It is a local martingale with quadratic variation $[X, X](t) = \int_0^t f'(s)ds = f(t)$. Thus $\tau_t = f^{(-1)}(t)$, the inverse of f , and according to the Change of Time Theorem, the process $X(f^{(-1)}(t)) = \hat{B}(t)$ is a Brownian motion (with respect to $\mathcal{F}_{\tau_t}$), and

$$X(t) = \hat{B}(f(t)). \quad (7.50)$$

Thus from equations (7.49) and (7.50) we have

Theorem 7.38 *Let $f(t)$ be an adapted, positive, increasing, differentiable process, and*

$$dX(t) = \sqrt{f'(t)}dB(t). \quad (7.51)$$

Then the process $B(f(t))$ is a weak solution.

We can write equation (7.51) as follows: for a Brownian motion B , and a function f , there is a Brownian motion $\hat{B}$, such that

$$d\hat{B}(f(t)) = \sqrt{f'(t)}dB(t). \quad (7.52)$$

In the case of non-random change of time in Brownian motion $B(f(t))$, it is easy to check directly that $M(t) = B(f(t))$ is a martingale (with respect to the filtration $\mathcal{F}_{f(t)}$). The quadratic variation of $B(f(t))$ is $[M, M](t) = [B(f), B(f)](t) = f(t)$, was calculated directly in the Example 4.24, equation (4.63).

Example 7.16: (Ornstein-Uhlenbeck process as Time Changed Brownian motion) With $f(t) = \sigma^2(e^{2\alpha t} - 1)/(2\alpha)$, the process $B(\sigma^2(e^{2\alpha t} - 1)/(2\alpha))$ is a weak solution to the SDE

$$dX(t) = \sigma e^{\alpha t} dB(t).$$

Consider $U(t) = e^{-\alpha t} X(t)$. Integrating by parts, $U(t)$ satisfies

$$dU(t) = -\alpha U(t)dt + \sigma dB(t). \quad (7.53)$$

Recall that the solution to this SDE is given by (5.13). Thus $U(t)$ is an Ornstein-Uhlenbeck process (see Example 5.6). Thus an Ornstein-Uhlenbeck process has representation

$$U(t) = e^{-\alpha t} B(\sigma^2(e^{2\alpha t} - 1)/(2\alpha)). \quad (7.54)$$

To have $U(0) = x$, take $B(t)$ to be a Brownian motion started at x . Note that in equations (7.53) and (7.54), $B(t)$ denotes different Brownian motions.

Next we construct a weak solution to the SDEs of the form

$$dX(t) = \sigma(X(t))dB(t)$$

with $\sigma(x) > 0$ such that

$$G(t) = \int_0^t \frac{ds}{\sigma^2(B(s))}$$

is finite for finite t , and increases to infinity, $\int_0^\infty \frac{ds}{\sigma^2(B(s))} = \infty$ almost surely. Then $G(t)$ is adapted, continuous and strictly increasing to $G(\infty) = \infty$. Therefore it has inverse

$$\tau_t = G^{(-1)}(t). \quad (7.55)$$

Note that for each fixed t , τ_t is a stopping time, as it is the first time the process $G(s)$ hits t , and that τ_t is increasing.

Theorem 7.39 *The process $X(t) = B(\tau_t)$ is a weak solution to the SDE*

$$dX(t) = \sigma(X(t))dB(t). \quad (7.56)$$

PROOF: $X(t) = B(\tau_t) = B(G^{(-1)}(t))$. Using equation (7.52) with $f = G^{(-1)}$, we obtain

$$dB(G^{(-1)}(t)) = \sqrt{(G^{(-1)})'(t)} d\hat{B}(t).$$

$$(G^{(-1)})'(t) = \frac{1}{G'(G^{(-1)}(t))} = \frac{1}{1/\sigma^2(B(G^{(-1)}(t)))} = \sigma^2(B(\tau_t)). \quad (7.57)$$

Thus we obtain

$$dB(\tau_t) = \sigma^2(B(\tau_t))d\hat{B}(t),$$

and the result is proved. $\square$

Another proof, by considering the martingale problem, is given next. Note that (7.57) gives $d\tau_t = \sigma^2(B(\tau_t))dt$.

PROOF: The diffusion operator for the SDE (7.56) is given by

$Lf(x) = \frac{1}{2}\sigma^2(x)f''(x)$. We show that $X(t) = B(\tau_t)$ is a solution to the martingale problem for L . Indeed, we know (see Example 5.17) that for any twice continuously differentiable function f vanishing outside a compact interval, the process $M(t) = f(B(t)) - \int_0^t \frac{1}{2}f''(B(s))ds$ is a martingale. Since τ_t are increasing stopping times it can be shown (by using Optional Stopping Theorem 7.18) that the process $M(\tau_t)$ is also a martingale, that is, $f(B(\tau_t)) - \int_0^{\tau_t} \frac{1}{2}f''(B(s))ds$ is a martingale. Now perform the change of variable $s = \tau_u$, and observe from (7.57) that $d\tau_t = \sigma^2(B(\tau_t))dt$, to obtain that the process $f(B(\tau_t)) - \int_0^t \frac{1}{2}\sigma^2(B(\tau_u))f''(B(\tau_u))du$ is a martingale. But since $X(t) = B(\tau_t)$, this being the same as $f(X(t)) - \int_0^t \frac{1}{2}\sigma^2(X(u))f''(X(u))du$ is a martingale, and $X(t)$ solves the martingale problem for L . $\square$

An application of Theorem 7.37 gives a result on uniqueness of the solution of SDE (7.56). This result is weaker than Theorem 6.13 of Engelbert-Schmidt.

Theorem 7.40 *Let $\sigma(x)$ be a positive function bounded away from zero, $\sigma(x) \geq \delta > 0$. Then the stochastic differential equation (7.56) has a unique weak solution.*

PROOF: Let $X(t)$ be a weak solution to (7.56). Then $X(t)$ is a local martingale and there is a Brownian motion $\beta(t)$, such that $X(t) = \beta([X, X](t))$. Now,

$$[X, X](t) = \int_0^t \sigma^2(X(s))ds = \int_0^t \sigma^2(\beta([X, X](s)))ds.$$

Thus $[X, X](t)$ is a solution to the ordinary differential equation (ODE) $da(t) = \sigma^2(\beta(a(t)))dt$. Since the solution to this ODE is unique, the solution to (7.56) is unique. $\square$

A more general change of time is done for the stochastic differential equation

$$dX(t) = \mu(X(t))dt + \sigma(X(t))dB(t). \quad (7.58)$$

Let $g(x)$ be a positive function for which $G(t) = \int_0^t g(X(s))ds$ is finite for finite t and increases to infinity almost surely. Define $\tau_t = G^{(-1)}(t)$.

Theorem 7.41 *Let $X(t)$ be a solution to (7.58) and define $Y(t) = X(\tau_t)$. Then $Y(t)$ is a weak solution to the stochastic differential equation*

$$dY(t) = \frac{\mu(Y(t))}{g(Y(t))}dt + \frac{\sigma(Y(t))}{\sqrt{g(Y(t))}}dB(t), \quad \text{with } Y(0) = X(0). \quad (7.59)$$

One can use the change of time on an interval $[0, T]$, for a stopping time T .

Example 7.17: (Lamperti's Change of Time)

Let $X(t)$ satisfy the SDE (Feller's branching diffusion)

$$dX(t) = \mu X(t)dt + \sigma \sqrt{X(t)}dB(t), \quad X(0) = x > 0, \quad (7.60)$$

with positive constants μ and σ . Lamperti's change of time is $G(t) = \int_0^t X(s)ds$. Here $g(x) = x$. Then $Y(t) = X(\tau_t)$ satisfies the SDE

$$\begin{aligned} dY(t) &= \frac{\mu Y(t)}{Y(t)}dt + \frac{\sigma \sqrt{Y(t)}}{\sqrt{Y(t)}}dB(t), \\ &= \mu dt + \sigma dB(t) \quad \text{with } Y(0) = x, \end{aligned}$$

and

$$Y(t) = x + \mu t + \sigma B(t). \quad (7.61)$$

In other words, with a random change of time, the Branching diffusion is a Brownian motion with drift. At the (random) point where $G(t)$ stops increasing its inverse τ_t , defined as the right-inverse $\tau_t = \inf\{s : G(s) = t\}$, also remains the same. This happens at the point of time when $X(t) = 0$. It can be seen that once the process is at zero, it stays at 0 forever. Let $T = \inf\{t : X(t) = 0\}$. T is a stopping time, and $Y(t)$ is the Brownian motion stopped at that time.

The other direction is also true, a Branching diffusion can be obtained from a Brownian motion with drift. Let $Y(t)$ satisfy (7.61), and let $T = \inf\{t : Y(t) = 0\}$. $Y(t) > 0$ for $t \leq T$. Define

$$G(t) = \int_0^{t \wedge T} \frac{1}{Y(s)}ds,$$

and let τ_t be the inverse of G , which is well defined on $[0, T)$. Then $X(t) = Y(\tau_t)$ satisfies the SDE (7.60) stopped when it hits zero.

Remark 7.4: Any solution to an SDE with time independent coefficients can be obtained from Brownian motion by using change of variables and random time change (Gihman and Skorohod (1972), p.113).

There are three main methods used for solving SDEs: change of state space, that is, change of variable (Itô's formula), change of time and change of measure. We have seen examples of SDEs solved by using change of variables, and change of time. The change of measure approach will be covered later.

Notes. Material for this chapter is based on Protter (1992), Rogers and Williams (1990), Gihman and Skorohod (1972), Liptser and Shiriyayev (1977), (1989), Revuz and Yor (1998).

7.9 Exercises

Exercise 7.1: Let $M(t)$ be an $\mathcal{F}_t$ -martingale and denote its natural filtration by $\mathcal{G}_t$. Show that $M(t)$ is a $\mathcal{G}_t$ -martingale.

Exercise 7.2: Show that an increasing integrable process is a submartingale.

Exercise 7.3: Show that if $X(t)$ is a submartingale and g is a non-decreasing convex function such that $E|g(X(t))| < \infty$, then $g(X(t))$ is a submartingale.

Exercise 7.4: Show that $M(t)$ is a square integrable martingale if and only if $M(t) = E(Y|\mathcal{F}_t)$, where Y is square integrable, $E(Y^2) < \infty$.

Exercise 7.5: (Expected exit time of Brownian motion from (a, b) .)

Let $B(t)$ be a Brownian motion started at $x \in (a, b)$, and $\tau = \inf\{t : B(t) = a \text{ or } b\}$. By stopping the martingale $M(t) = B(t)^2 - t$, show that $E_x(\tau) = (x - a)(b - x)$.

Exercise 7.6: Find the probability of $B(t) - t/2$ reaching a before it reaches b when started at x , $a < x < b$. Hint: use the exponential martingale $M(t) = e^{B(t) - t/2}$.

Exercise 7.7: Find the expected length of the game in Gambler's ruin, when

- betting is done on a fair coin
- betting is done on a biased coin

Exercise 7.8: Give the probability of ruin when playing a game of chance against an infinitely rich opponent (with initial capital $b \rightarrow \infty$).

Exercise 7.9: (Ruin Probability in Insurance) A Discrete Time Risk Model for the surplus U_n of an insurance company at the end of year n , $n = 1, 2, \dots$ is given by $U_n = U_0 + cn - \sum_{k=1}^n X_k$, where c is the total annual premium, X_k is the total (aggregate) claim in year k . The time of ruin T is the first time when the surplus becomes negative, $T = \min\{n : U_n < 0\}$, with $T = \infty$ if $U_n \geq 0$ for all n . Assume that $\{X_k, k = 1, 2, \dots\}$ are i.i.d. random variables, and there exists a constant $R > 0$ such that $E(e^{-R(c-X_1)}) = 1$. Show that for all n , $P_x(T \leq n) \leq e^{-Rx}$, where $U_0 = x$ the initial funds, and the ruin probability $P_x(T < \infty) \leq e^{-Rx}$. Hint: show that $M_n = e^{-RU_n}$ is a martingale, and use the Optional Stopping Theorem.

Exercise 7.10: (Ruin Probability in Insurance continued) Find the bound on the ruin probability when the aggregate claims have $N(\mu, \sigma^2)$ distribution. Give the initial amount x required to keep the ruin probability below level α .

Exercise 7.11: Let $B(t)$ be a Brownian motion starting at zero and T be the first exit time from $(-1, 1)$, that is, the first time when $|B|$ takes value 1. Use Davis' inequality to show that $E(\sqrt{T}) < \infty$.

Exercise 7.12: Let $B(t)$ be a Brownian motion, $X(t) = \int_0^t \text{sign}(B(s))dB(s)$. Show that X is also a Brownian motion.

Exercise 7.13: Let $M(t) = \int_0^t e^s dB(s)$. Find $g(t)$ such that $M(g(t))$ is a Brownian motion.

Exercise 7.14: Let $B(t)$ be a Brownian motion. Give an SDE for $e^{-\alpha t}B(e^{2\alpha t})$.

Exercise 7.15: Prove the change of time result in SDEs, Theorem 7.41.

Exercise 7.16: Let $X(t)$ satisfy SDE $dX(t) = \mu(t)dt + \sigma(t)dB(t)$ on $[0, T]$. Show that $X(t)$ is a local martingale if and only if $\mu(t) = 0$ a.e.

Exercise 7.17: $f(x, t)$ is differentiable in t and twice in x . It is known that $X(t) = f(B(t), t)$ is of finite variation. Show that f is a function of t alone.

Exercise 7.18: Let $Y(t) = \int_0^t B(s)dB(s)$ and $W(t) = \int_0^t \text{sign}(B(s))dB(s)$. Show that $dY(t) = \sqrt{t + 2Y(t)}dW(t)$. Show uniqueness of the weak solution of the above SDE.